Revised Project Description

For my course project, I plan to investigate how human–AI collaboration can be designed to support trust, safety, and usability in autonomous vehicles (AVs). This builds on my interests in accessibility, human–AI collaboration, and AVs. Although AVs are becoming more technologically advanced, they still face scrutiny based on concerns about trust, safety, and transparency. Unlike conventional driving, AVs shift people into supervisory, collaborative, or even passive roles depending on the level of automation, raising questions about what "good" collaboration looks like. Specifically, collaboration may involve how AVs explain decisions, request input, or transfer control back to the human driver. My aim is to explore how everyday users experience collaboration with AVs and how different interaction modalities (voice, gesture, visual cues, haptics) may support or hinder trust and sense of control. In doing so, this study will generate insights into how design strategies shape human–AI teaming in the AV context, with attention to accessibility for a broad range of users.

Revised Research Questions

RQ1: How do different modes of human–AI communication (e.g., visual, auditory, or haptic) affect user trust and sense of control in autonomous vehicles?

RQ2: What kinds of signals or triggers do users prefer for safe and seamless transitions of control between the AI system and the human driver?

Participants and sampling

Target group: Adults (18+) who have at least some driving experience with AVs.

Recruitment strategy: Convenience sampling through classmates, friends and word of mouth.

Sample size goal: 2–3 semi-structured interviews of 30 minutes each.

Data Collection Method

Approach: Semi-structured interviews lasting **30–60 minutes**.

Format: Conducted via Zoom or in person.

Interview Guide

1. Can you tell me about your experience with driving an autonomous vehicle? Probe: What excites or worries you about AVs?

2. When you ride the AV, what are the ways you receive information from the AV and what are the information communicated to you?
3. What kinds of communication (e.g., voice, screens, alerts) make you feel most comfortable?
4. Have you experienced a driving situation where the AV communicated with you in a way that felt trustworthy or untrustworthy? If yes, What made it trustworthy or untrustworthy?
5. What experiences have you had with your AV that made you feel less in control?
6. In terms of control, since the AV is self-driving, what scenarios would you want to take control and how do you think the AV should hand control back to you? Probe: Would you prefer a sound, a visual cue, or another kind of signal?
7. In an emergency, what would be your preference for the AV to alert you to take over?
8. Would you want an instantaneous transfer of control, or would you like to be prompted first, and how much warning time would you want before the AV hands over control?
9. Do you think certain modes of communication (voice, gesture, haptics, visuals) would be more accessible or less accessible for you personally? Why? Probe: Are there people you know who might need different designs, e.g. older adults, people who are hard of hearing, etc.?
10. If you could give AV designers one piece of advice about communication and collaboration, what would it be?